# QFLPN State Representation versus Matrix Product States: Compression, Fidelity, and Entanglement-Regime Selection

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**Abstract**

Quantum-Fuzzy Logical Petri Nets (QFLPN) combine discrete event structure, fuzzy activation, and quantum-state evolution. A representation layer is therefore required to store and evolve the quantum state produced by the QFLPN semantics. This paper studies a precise representation question: when can a Matrix Product State (MPS) replace an exact state vector while preserving the represented state and reducing the declared storage payload? Six controlled state families are evaluated: product, GHZ, one-dimensional cluster, random circuits of depth 2 and 4, and Haar-random states. For each family, normalized exact states are decomposed by sequential tensor-train singular-value decomposition (TT-SVD) with tolerance $10^{-8}$ for $q \in \{6, 8, 10, 12, 14\}$, producing 30 instances. Maximum bond dimension, payload memory, compression ratio, fidelity, discarded weight, and TT-SVD time are recorded. At $q = 14$, product, GHZ, cluster, random-depth-2, and random-depth-4 states retain storage advantages of 585.14×, 157.54×, 157.54×, 81.92×, and 7.39×, respectively. The Haar-random state reaches bond dimension 128 and a compression ratio of 0.3750, so its MPS payload is larger than the exact-vector payload under the declared accounting. The resulting evidence supports a conditional representation rule based on bond-dimension growth, fidelity, and storage advantage rather than qubit count alone. The study does not claim universal runtime superiority, hardware-independent scalability, or a universal advantage of MPS over exact state vectors.

**Keywords:** quantum-fuzzy logical Petri nets; matrix product states; tensor networks; TT-SVD; Schmidt rank; entanglement; state compression; fidelity; representation selection.

## 1 Introduction

QFLPN is a hybrid formalism in which a discrete Petri-net structure determines admissible transitions, fuzzy variables determine activation strengths, and quantum operators determine the evolution of a quantum state. Once a state is produced, however, its numerical representation becomes a separate computational design choice. An exact $q$-qubit state vector has dimension $2^q$ and is written as

$$|\psi\rangle = \sum_{j=0}^{2^q-1} c_j |j\rangle, \qquad \sum_{j=0}^{2^q-1} |c_j|^2 = 1. \tag{1}$$

With complex double precision, the amplitude payload alone requires approximately $16\,2^q$ bytes. This exponential dependence motivates compressed representations, but compression is not automatically beneficial: a compressed representation can require more storage when the state has large effective Schmidt rank, and conversion itself has a non-negligible cost.

An MPS represents the same state as a sequence of local tensors,

$$|\psi\rangle = \sum_{i_1,\dots,i_q} A^{[1]}_{i_1} A^{[2]}_{i_2} \cdots A^{[q]}_{i_q} |i_1 \cdots i_q\rangle, \tag{2}$$

where the internal tensor dimensions are the bond dimensions. For a bipartition after site $k$, the maximum required bond dimension equals the Schmidt rank of the state across that cut when the representation is exact. The useful scientific question is therefore not whether MPS is globally superior, but which state structures create a favorable compression regime.

The central research question is: under what observable state-structure conditions can an MPS representation replace the exact vector while meeting a declared fidelity criterion and reducing the representation payload? A second question is whether states with the same qubit count can occupy fundamentally different representation regimes. The experiment is designed to answer these questions using controlled state families and a fixed decomposition tolerance.

The contribution is deliberately representation-level. The QFLPN semantics are treated as the source of the quantum state, while MPS is evaluated as a numerical representation of that state. The paper therefore does not introduce a new QFLPN transition rule, and it does not infer physical or computational superiority from compression alone.

## 2 Scientific Context and Tensor-Network Background

Matrix Product States are a standard one-dimensional tensor-network representation and are closely connected to Schmidt decompositions, density-matrix renormalization, and low-rank structure [1, 2, 3]. Their computational value arises when the state admits small effective bond dimensions under the selected site ordering. This connection is also reflected in the classical simulation of weakly entangled quantum systems, where the relevant computational resources depend strongly on entanglement rather than on qubit count alone [4, 5]. Their computational value arises when the state admits small effective bond dimensions under the selected site ordering. The present study adopts this established representation as a baseline for a controlled QFLPN representation comparison.

For a normalized pure state and a bipartition after site $k$, the Schmidt decomposition is

$$|\psi\rangle = \sum_{r=1}^{R_k} s^{(k)}_r |u^{(k)}_r\rangle \otimes |v^{(k)}_r\rangle, \qquad s^{(k)}_1 \geq s^{(k)}_2 \geq \cdots \geq 0. \tag{3}$$

The normalization condition is $\sum_r (s^{(k)}_r)^2 = 1$. If only the first $D_k$ coefficients are retained, the discarded weight at that cut is

$$\varepsilon_k = \sum_{r>D_k} (s^{(k)}_r)^2. \tag{4}$$

TT-SVD applies these low-rank decompositions sequentially. The tensor-train formulation provides the numerical foundation for this decomposition and for subsequent rounding and low-rank operations [9]. The resulting bond dimensions provide a direct structural diagnostic, while the reconstructed-state fidelity determines whether the compression is acceptable for the stated purpose.

The distinction between structural complexity and approximation quality is essential. Tensor-network efficiency is fundamentally tied to entanglement structure; area-law results provide an important theoretical explanation for why compact tensor-network representations can occur for broad classes of low-energy states, while not implying compactness for arbitrary states [7]. A low bond dimension indicates a compact representation, but it is not by itself a certificate that the reconstructed state is sufficiently accurate. Conversely, fidelity close to unity does not imply that the representation is memory-efficient. The experiment therefore records both quantities and treats them as separate evidence channels.

## 3 QFLPN Representation Interface

Let

$$\mathcal{N}_{\mathrm{QFLPN}} = (P, T, E, \mathcal{H}, \mathcal{F}) \tag{5}$$

denote a QFLPN instance with global Hilbert space

$$\mathcal{H}_{\mathrm{glob}} = \bigotimes_{p \in P} \mathcal{H}_p. \tag{6}$$

A fuzzy transition activation may be expressed as

$$\lambda_t = g(\mu_1(t), \ldots, \mu_q(t)), \qquad 0 \leq \lambda_t \leq 1, \tag{7}$$

with rotation parameter

$$\theta_t = 2 \arcsin \sqrt{\lambda_t}. \tag{8}$$

At the representation boundary, the semantic flow is

$$\mathcal{M} \longrightarrow \lambda_t \longrightarrow \theta_t \longrightarrow \rho' \longrightarrow \mathfrak{R}(\rho'), \tag{9}$$

where $\mathfrak{R}$ is the numerical representation map. This paper compares two realizations of $\mathfrak{R}$: direct storage as an exact state vector and sequential TT-SVD storage as an MPS. The semantic state itself is unchanged by the choice of representation.

For normalized pure states $|\psi\rangle$ and $|\widetilde{\psi}\rangle$, fidelity is

$$F = |\langle \psi | \widetilde{\psi} \rangle|^2. \tag{10}$$

The experiment reports $F$ directly rather than imposing a universal application threshold. This keeps the representation evidence independent from any particular downstream control or decision requirement.

## 4 Equivalence Protocol and Storage Accounting

A valid representation comparison must keep the semantic state, precision convention, tensor ordering, and error metric fixed. The exact state vector is the reference. TT-SVD is then applied to that same normalized state. The reconstructed MPS is contracted back to a state vector for fidelity evaluation. The resulting difference is therefore attributable to the representation transformation and truncation, not to a change in the underlying state definition.

For complex double payloads, the exact vector storage is

$$S_{\mathrm{vec}}(q) = 16\,2^q \quad \text{bytes.} \tag{11}$$

For an open-boundary binary MPS with physical dimension $d = 2$ and boundary dimensions $D_0 = D_q = 1$,

$$S_{\mathrm{MPS}} = 16 \sum_{k=1}^{q} d D_{k-1} D_k = 32 \sum_{k=1}^{q} D_{k-1} D_k \quad \text{bytes.} \tag{12}$$

The accounting excludes object headers, allocator overhead, metadata, padding, and implementation-specific tensor containers. It therefore measures payload size rather than complete process memory.

The compression ratio is

$$C = \frac{S_{\mathrm{vec}}}{S_{\mathrm{MPS}}}. \tag{13}$$

Values $C > 1$ indicate a smaller MPS payload, while $C < 1$ indicates that the MPS payload is larger under the declared accounting. TT-SVD wall-clock time is reported separately because representation size and construction cost are different performance dimensions.

## 5 Analytical Scaling Regimes

Suppose an open-boundary binary MPS has an internal bond bound $D_k \leq D$. Then

$$S_{\mathrm{MPS}} = 32 \sum_{k=1}^{q} D_{k-1} D_k \leq 32 q D^2. \tag{14}$$

Consequently,

$$C \geq \frac{2^q}{2qD^2}. \tag{15}$$

This inequality is an analytical scaling statement: a genuinely bounded-bond family eventually has an exponentially increasing payload advantage over an exact vector. It does not imply that every quantum state has bounded bond dimension.

At the opposite extreme, an even-$q$ state with a full-rank central bipartition can require Schmidt rank $2^{q/2}$. An exact MPS must then have $D_{\max} \geq 2^{q/2}$. This regime removes the bounded-bond assumption and explains why highly entangled states can lose the storage advantage. The measured Haar-random sequence is consistent with this high-rank regime for the tested sizes, but the analytical statement is independent of the measurements.

The resulting conceptual boundary is therefore structural. This interpretation is consistent with the broader tensor-network literature, in which different network geometries and algorithms are selected according to the entanglement and interaction structure of the target problem [8, 12]. Qubit count determines the size of the exact state space, but bond-dimension growth determines whether a tensor representation can exploit structure within that space.

## 6 Experimental Design

Six state families were selected to span qualitatively different structural regimes. Product states provide a separable reference. GHZ states exhibit global correlation while retaining small Schmidt rank. One-dimensional cluster states provide a graph-state family with bounded bond dimension under the selected ordering. Random circuits of depth 2 and 4 progressively increase entangling structure. Haar-random states provide a high-entanglement reference with near-maximal bipartite rank. This choice is also motivated by the known sensitivity of classical quantum-circuit simulation to circuit structure, graph width, contraction paths, and entanglement growth [10, 13, 14].

The experiment evaluates $q \in \{6, 8, 10, 12, 14\}$ for each family, giving 30 instances. Randomized constructions use seed 20260908. Each state is normalized before decomposition. TT-SVD uses tolerance $10^{-8}$. The recorded metrics are maximum bond dimension, exact-vector payload, MPS payload, compression ratio, fidelity, discarded weight, and TT-SVD wall-clock time.

Table 1: Declared experimental protocol.

| Parameter | Value |
|---|---|
| State families | Product; GHZ; Cluster; Random depth 2; Random depth 4; Haar random |
| Qubit counts | $q \in \{6, 8, 10, 12, 14\}$ |
| Number of instances | 30 |
| Reference representation | normalized exact state vector |
| Compression method | sequential TT-SVD |
| Tolerance | $10^{-8}$ |
| Random seed | 20260908 |
| Payload precision | complex double |
| Fidelity | $\lvert\langle\psi\vert\widetilde{\psi}\rangle\rvert^2$ |

## 7 Results: Compression Across State Families

Figure 1 summarizes the measured compression ratios. Product, GHZ, and cluster states remain strongly compressible as $q$ increases. Random depth 2 also remains favorable. Random depth 4 approaches a less favorable regime because its bond dimension has already increased substantially. Haar-random states remain below unity throughout the measured range.

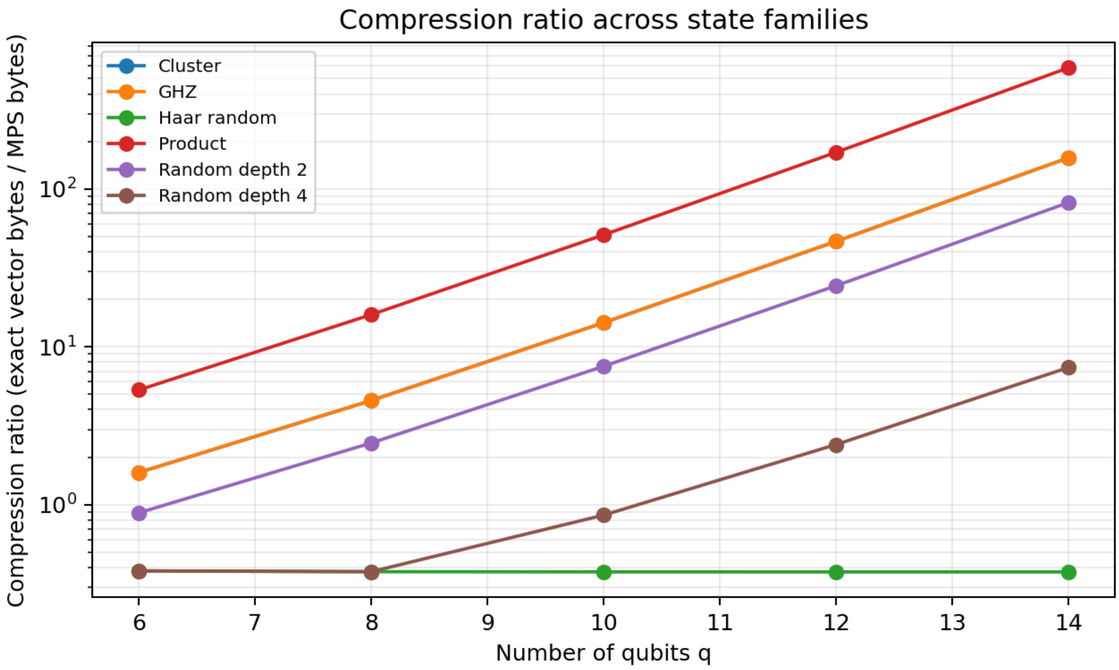


Figure 1: Measured compression ratio versus qubit count for the six state families.

The measurements show that equal qubit count does not determine equal representation cost. At $q = 14$, the product family reaches $C = 585.142857$, while the Haar-random family reaches $C = 0.375023$. The two states therefore differ by more than three orders of magnitude in compression ratio despite having the same Hilbert-space dimension.

This is the first central empirical result: representation selection must depend on state structure, not only on $q$.

## 8 Results: Bond-Dimension Growth

The maximum bond dimension is the structural variable that explains the observed storage behavior. Figure 2 shows the measured growth. Product states remain at 1, GHZ and cluster states at 2, and random depth 2 at 4 over the tested sizes. Random depth 4 reaches 16. Haar-random states follow 8, 16, 32, 64, and 128 as $q$ increases from 6 to 14.

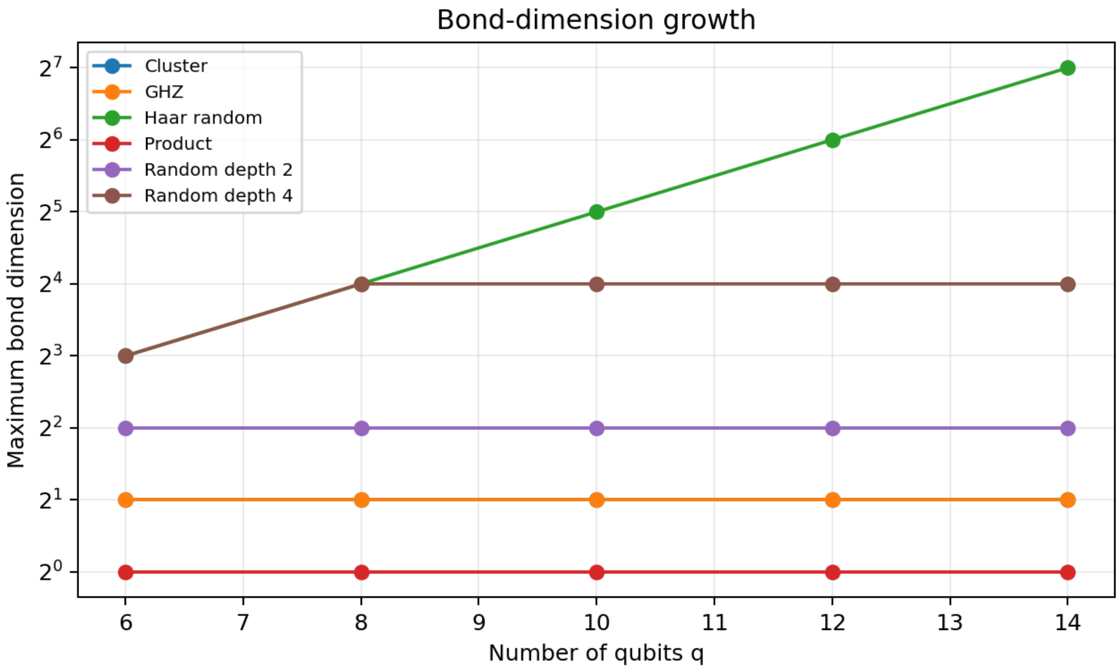


Figure 2: Maximum measured MPS bond dimension versus qubit count.

The data support a direct interpretation: bounded or slowly growing bond dimensions yield compact MPS payloads, while exponential bond growth removes the advantage. This interpretation is structural and does not depend on any assumption that MPS is universally efficient.

# 9 Results: The $q = 14$ Representation Boundary

Table 2 gives the complete $q = 14$ cross-family snapshot.

Table 2: Measured $q = 14$ representation results.

| Family | $D_{\max}$ | MPS bytes | $C$ | $F$ | TT-SVD (s) |
|---|---|---|---|---|---|
| Product | 1 | 448 | 585.143 | 1.000000000000 | 0.00202209 |
| GHZ | 2 | 1664 | 157.538 | 1.000000000000 | 0.00271267 |
| Cluster | 2 | 1664 | 157.538 | 1.000000000000 | 0.0172741 |
| Random depth 2 | 4 | 3200 | 81.920 | 1.000000000000 | 0.00467951 |
| Random depth 4 | 16 | 35456 | 7.39350 | 1.000000000000 | 0.0102184 |
| Haar random | 128 | 699008 | 0.375023 | 1.000000000000 | 0.0614909 |

The Haar-random case is the only tested family at $q = 14$ for which the MPS payload exceeds the exact-vector payload. The crossover is not caused by reduced fidelity: the measured fidelities remain numerically indistinguishable from unity. Instead, the cause is the high bond dimension required by the state structure.

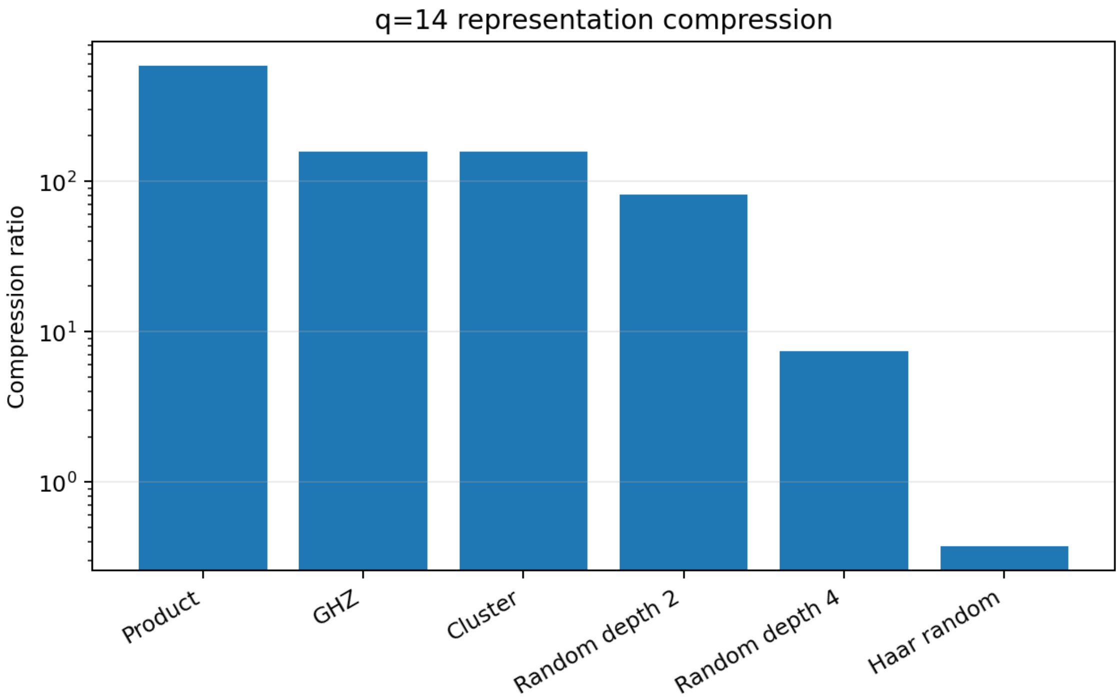


Figure 3: Compression ratio at $q = 14$. The Haar-random case lies below unity.

## 10 Complete 30-Instance Benchmark

The complete benchmark is reported at row level so that every aggregate statement can be traced to the underlying measurements. The first table contains structural and accuracy metrics; the second contains the storage payloads and discarded weights. The two tables use the same 30 instances and preserve the values recorded in the accompanying CSV artifact.

Table 3: Complete benchmark: structural, compression, fidelity, and conversion metrics.

| Family | $q$ | $D_{\max}$ | $C$ | $F$ | TT-SVD s |
|---|---|---|---|---|---|
| Product | 6 | 1 | 5.33333 | 1 | 0.00126558 |
| Product | 8 | 1 | 16 | 1 | 0.000182645 |
| Product | 10 | 1 | 51.2 | 1 | 0.000253811 |
| Product | 12 | 1 | 170.667 | 1 | 0.000432855 |
| Product | 14 | 1 | 585.143 | 1 | 0.00202209 |
| GHZ | 6 | 2 | 1.6 | 1 | 0.000180418 |
| GHZ | 8 | 2 | 4.57143 | 1 | 0.000199263 |
| GHZ | 10 | 2 | 14.2222 | 1 | 0.000351912 |
| GHZ | 12 | 2 | 46.5455 | 1 | 0.000582971 |
| GHZ | 14 | 2 | 157.538 | 1 | 0.00271267 |
| Cluster | 6 | 2 | 1.6 | 1 | 0.000241383 |
| Cluster | 8 | 2 | 4.57143 | 1 | 0.000285084 |
| Cluster | 10 | 2 | 14.2222 | 1 | 0.000607495 |
| Cluster | 12 | 2 | 46.5455 | 1 | 0.000802774 |
| Cluster | 14 | 2 | 157.538 | 1 | 0.0172741 |
| Random depth 2 | 6 | 4 | 0.888889 | 1 | 0.000276832 |
| Random depth 2 | 8 | 4 | 2.46154 | 1 | 0.000489215 |
| Random depth 2 | 10 | 4 | 7.52941 | 1 | 0.000599411 |
| Random depth 2 | 12 | 4 | 24.381 | 1 | 0.00133135 |
| Random depth 2 | 14 | 4 | 81.92 | 1 | 0.00467951 |
| Random depth 4 | 6 | 8 | 0.380952 | 1 | 0.000255317 |
| Random depth 4 | 8 | 16 | 0.376471 | 1 | 0.00043689 |
| Random depth 4 | 10 | 16 | 0.85906 | 1 | 0.00100669 |
| Random depth 4 | 12 | 16 | 2.40376 | 1 | 0.00264687 |
| Random depth 4 | 14 | 16 | 7.3935 | 1 | 0.0102184 |

| Family | $q$ | $D_{\max}$ | $C$ | $F$ | TT-SVD s |
|---|---|---|---|---|---|
| Haar random | 6 | 8 | 0.380952 | 1 | 0.000264912 |
| Haar random | 8 | 16 | 0.376471 | 1 | 0.000394456 |
| Haar random | 10 | 32 | 0.375367 | 1 | 0.00109158 |
| Haar random | 12 | 64 | 0.375092 | 1 | 0.00366113 |
| Haar random | 14 | 128 | 0.375023 | 1 | 0.0614909 |

Table 4: Complete benchmark: MPS payload, exact-vector payload, and discarded weight.

| Family | $q$ | MPS bytes | Vector bytes | $\varepsilon$ |
|---|---|---|---|---|
| Product | 6 | 192 | 1024 | 3.879e-31 |
| Product | 8 | 256 | 4096 | 4.609e-31 |
| Product | 10 | 320 | 16384 | 8.892e-29 |
| Product | 12 | 384 | 65536 | 6.513e-28 |
| Product | 14 | 448 | 262144 | 6.127e-27 |
| GHZ | 6 | 640 | 1024 | 0.000e+00 |
| GHZ | 8 | 896 | 4096 | 0.000e+00 |
| GHZ | 10 | 1152 | 16384 | 0.000e+00 |
| GHZ | 12 | 1408 | 65536 | 0.000e+00 |
| GHZ | 14 | 1664 | 262144 | 0.000e+00 |
| Cluster | 6 | 640 | 1024 | 5.447e-32 |
| Cluster | 8 | 896 | 4096 | 6.718e-31 |
| Cluster | 10 | 1152 | 16384 | 5.759e-30 |
| Cluster | 12 | 1408 | 65536 | 6.528e-29 |
| Cluster | 14 | 1664 | 262144 | 4.361e-27 |
| Random depth 2 | 6 | 1152 | 1024 | 3.578e-32 |
| Random depth 2 | 8 | 1664 | 4096 | 2.383e-31 |
| Random depth 2 | 10 | 2176 | 16384 | 1.172e-30 |
| Random depth 2 | 12 | 2688 | 65536 | 2.482e-30 |
| Random depth 2 | 14 | 3200 | 262144 | 5.132e-30 |
| Random depth 4 | 6 | 2688 | 1024 | 0.000e+00 |
| Random depth 4 | 8 | 10880 | 4096 | 0.000e+00 |
| Random depth 4 | 10 | 19072 | 16384 | 1.850e-31 |
| Random depth 4 | 12 | 27264 | 65536 | 6.710e-31 |
| Random depth 4 | 14 | 35456 | 262144 | 1.934e-30 |
| Haar random | 6 | 2688 | 1024 | 0.000e+00 |
| Haar random | 8 | 10880 | 4096 | 0.000e+00 |
| Haar random | 10 | 43648 | 16384 | 0.000e+00 |
| Haar random | 12 | 174720 | 65536 | 0.000e+00 |
| Haar random | 14 | 699008 | 262144 | 0.000e+00 |

The row-level data show that the compression ratio changes primarily with bond-dimension growth, while the fidelity remains at numerical unity for the selected tolerance. The storage tables also make the accounting distinction explicit: payload bytes are analytical representation costs and are not equivalent to complete process memory.

# 11 Fidelity and Truncation Evidence

The measured fidelity is effectively unity across all 30 instances. This result should be interpreted together with the discarded-weight column. For the structured families, the discarded weights are at numerical-roundoff scale. For the Haar-random family, the reported discarded weight is zero at the selected tolerance because the retained bond dimensions reach the required ranks for the tested states.

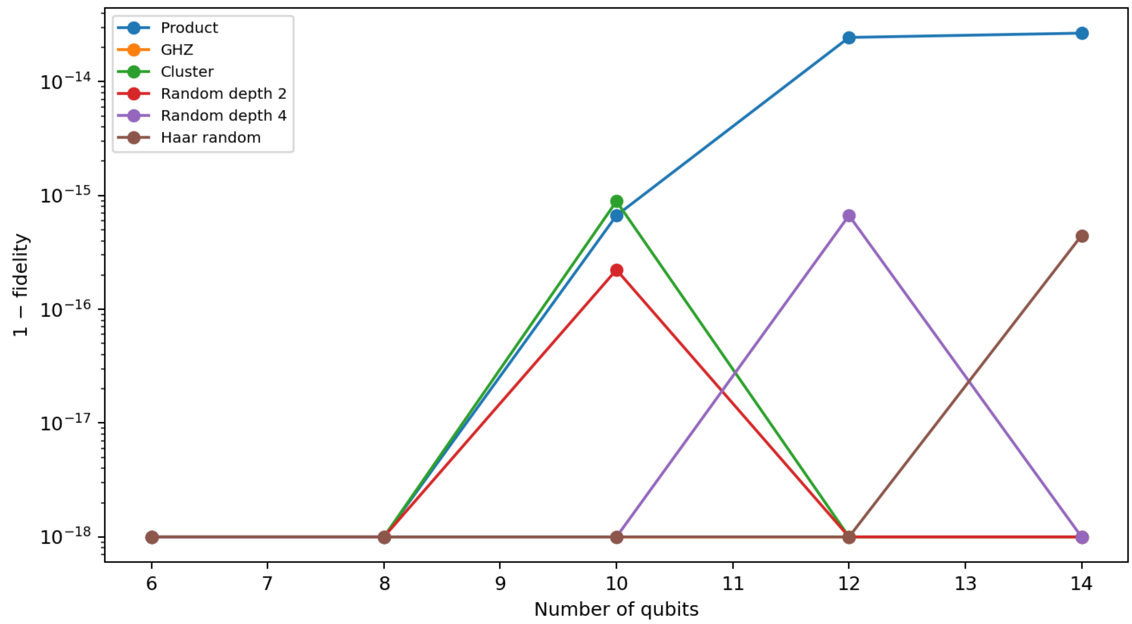


Figure 4: Measured infidelity across the benchmark. The selected tolerance preserves numerical fidelity for all tested instances.

The important methodological point is that fidelity and storage advantage are independent criteria. The Haar-random state demonstrates this directly: fidelity is effectively one, but the storage ratio is below one. Therefore, fidelity alone cannot justify choosing MPS.

## 12 Conversion Cost and Representation Trade-Off

TT-SVD time increases with state complexity. At $q = 14$, the product state requires approximately 2.02 ms, GHZ approximately 2.71 ms, random depth 2 approximately 4.68 ms, random depth 4 approximately 10.22 ms, and Haar random approximately 61.49 ms. The cluster state records approximately 17.27 ms despite its small final bond dimension, demonstrating that final representation size does not uniquely determine decomposition time.

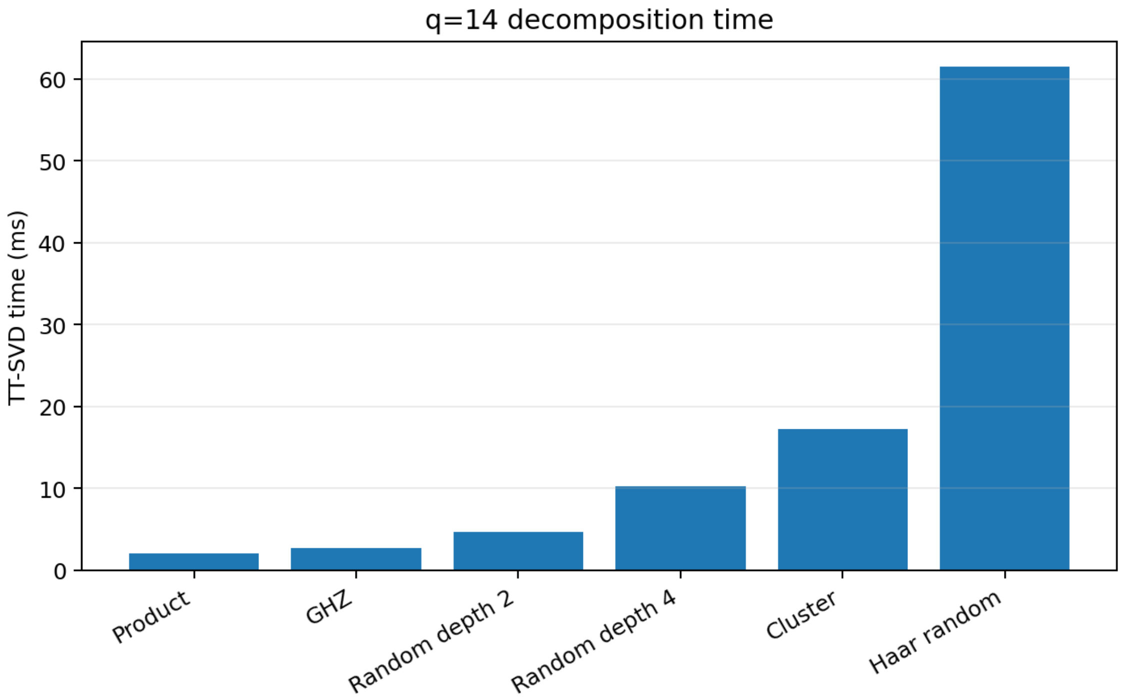


Figure 5: TT-SVD wall-clock time at $q = 14$. Conversion cost is distinct from final payload size.

This distinction matters for dynamic QFLPN workloads. If a state is converted to MPS once and then evolved for many operations, conversion cost may be amortized. This distinction becomes especially important for time-dependent simulations, where MPS-specific evolution algorithms introduce their own approximation and computational costs [15]. If states are repeatedly converted between representations, the transformation overhead can dominate. The present experiment measures the conversion step but does not benchmark a full time-dependent MPS evolution engine, so no universal runtime conclusion is drawn.

## 13 Representation Selection Criterion

The measurements motivate a simple evidence-based selection rule. Let $F_{\min}$ be an application-specific fidelity requirement and let $C$ be the payload compression ratio. An MPS representation is storage-admissible when

$$F \geq F_{\min} \quad \text{and} \quad C > 1. \tag{16}$$

This criterion is intentionally conservative. It does not select MPS solely because fidelity is high, and it does not select MPS solely because the bond dimension is small. A practical implementation can add a conversion-cost threshold and an evolution-cost model, but those quantities require workload-specific measurements.

Figure 6 visualizes the resulting representation regimes in the measured data. The structured families occupy the favorable region, while Haar-random states occupy the high-rank region.

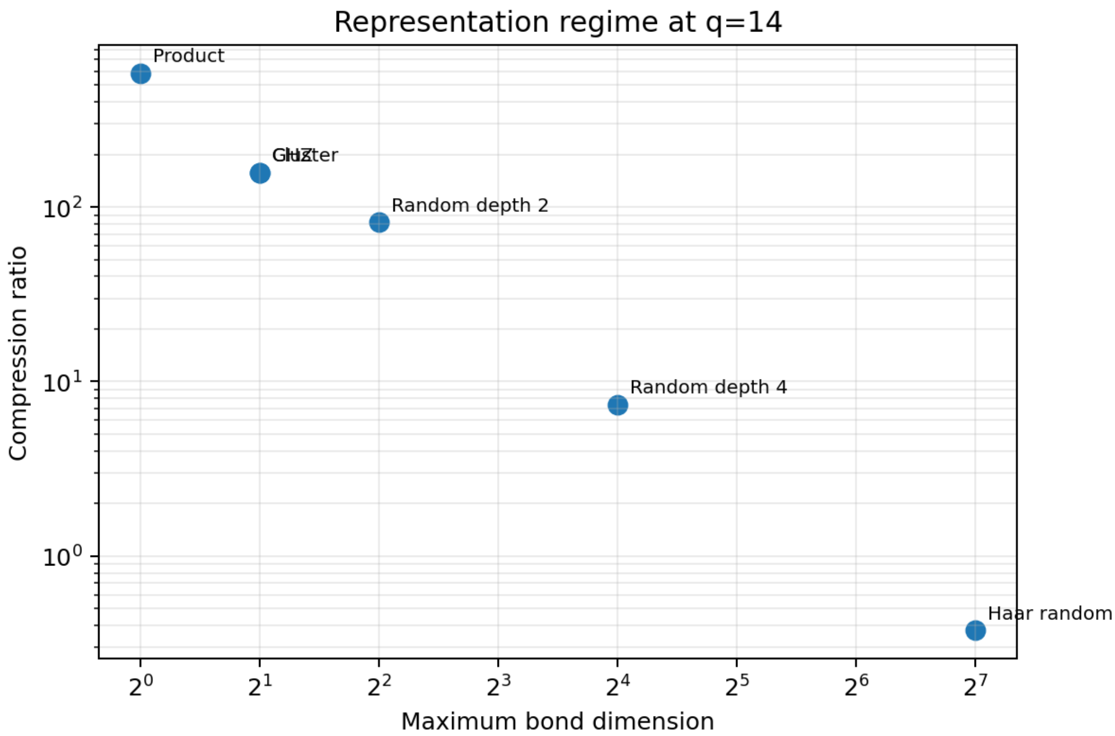


Figure 6: Measured representation regimes defined by compression and structural complexity.

## 14 Discussion

The main scientific conclusion is conditional rather than universal. MPS is strongly advantageous for the tested structured families because their bond dimensions remain small relative to the exponentially growing exact-vector dimension. The advantage weakens as entangling structure increases and disappears for the tested Haar-random family.

The result also clarifies the role of QFLPN. The Petri and fuzzy layers determine which transitions are activated and how the quantum state is semantically generated. Once the state exists, the representation layer should be selected according to its mathematical structure. Therefore, QFLPN does not imply MPS efficiency by construction. An application can legitimately require exact vectors, MPS, or another tensor-network representation depending on the generated state family.

A second important observation is that site ordering remains an external factor. The present benchmark fixes one ordering and does not optimize permutations. For graph states, random circuits, or application-specific interaction graphs, a different ordering can change the maximum bond dimension substantially. Consequently, the present measurements should not be interpreted as an ordering-independent optimum.

A third observation concerns precision. The payload equations use complex double precision, while the TT-SVD tolerance is $10^{-8}$. Different precisions or tolerances can move the representation boundary. A future benchmark should therefore report a joint sweep over precision, tolerance, ordering, and state structure rather than treating one configuration as universally optimal.

## 15 Limitations and Threats to Validity

The benchmark contains six state families and five qubit counts. It is therefore a controlled structural study, not an exhaustive characterization of all QFLPN states. The Haar-random family is a stress test for high entanglement, not a model of every physically relevant quantum workload.

The storage model counts tensor payloads only. Real implementations include metadata, object headers, memory alignment, temporary workspaces, and allocator behavior. The measured TT-SVD time is implementation-dependent and should not be generalized across hardware or software environments.

The experiment also evaluates representation conversion rather than full MPS operator evolution. Tensor-network contraction itself can have strongly problem-dependent complexity, and more general tensor-network geometries do not inherit the favorable one-dimensional MPS scaling automatically [11, 10, 13]. It therefore does not establish that MPS will provide a runtime speedup during long QFLPN trajectories. Such a claim requires matched evolution kernels, operator application benchmarks, and workload-level measurements.

Finally, the fidelity metric is defined for pure states. Open-system QFLPN models represented by density operators or quantum channels require an extension of the comparison protocol, for example through trace distance, state fidelity for mixed states, observable error, or channel-level metrics.

## 16 Reproducibility and Data Availability

The experiment is defined by the state-family matrix, qubit counts, deterministic seed, TT-SVD tolerance, payload equations, and fidelity definition stated in this paper. The complete numerical output is stored in the accompanying CSV file `P9_mps_statevector_benchmark.csv`. The figures are generated from the same benchmark data. The source document fixes the numerical table to the recorded measurements and uses no manually invented aggregate values.

For a fully external reproduction, the implementation should additionally record the exact software versions, BLAS/LAPACK backend, processor model, operating-system version, compiler/interpreter version, tensor ordering convention, and wall-clock measurement method. These environment details are not inferred here when they are not present in the supplied benchmark artifact.

## 17 Relationship to the Other QFLPN Studies

This paper is intentionally separated from the other studies in the series. Its primary object is the representation boundary between an exact state vector and an MPS. It does not redefine the QFLPN formalism, establish the four-qubit reference instance, benchmark CSR/SpMV execution, compare Taylor with Krylov–Arnoldi, develop adaptive Krylov integration, model memory degradation, analyze Lindblad robustness, or study spectral nonnormality.

The distinction is also important for overlap control with the dissertation. The QFLPN equations used here are interface definitions required to identify the represented state, while the original empirical contribution of this paper is the controlled 30-instance representation benchmark and the resulting structural selection criterion. Any final journal submission should cite the dissertation and any prior publications containing reused definitions or figures and should explicitly identify the new comparative evidence.

## 18 Conclusion

The study establishes a measurable boundary for choosing between exact state-vector and MPS representations in the tested QFLPN state families. Product, GHZ, cluster, random-depth-2, and random-depth-4 states retain substantial payload advantages as the number of qubits increases. Haar-random states do not: their bond dimension grows rapidly, and at $q = 14$ the MPS payload is approximately 2.67 times the exact-vector payload under the declared accounting.

The central conclusion is therefore not that MPS is universally superior. The evidence supports a representation rule based on three separable quantities: structural complexity through bond dimension, correctness through fidelity, and resource benefit through payload compression. This rule is compatible with QFLPN because it operates at the numerical representation layer without altering the underlying logical or quantum semantics.

The next scientifically necessary step is workload-level validation: matched QFLPN evolution benchmarks in which exact vectors and MPS are subjected to the same transition sequence, precision, fidelity criterion, operator family, ordering strategy, and hardware conditions. Only such an experiment can establish whether the storage regimes identified here translate into sustained computational advantage.

## Author Contributions

Adrian Roman: conceptualization of the representation comparison, mathematical formulation, experimental design, data analysis, manuscript preparation, and reproducibility specification.

## Data and Code Availability

The numerical benchmark accompanies the manuscript as the file `P9_mps_statevector_benchmark.csv`. The source records the complete 30-instance table and the declared accounting rules. Any future public code release should preserve the seed, tolerance, tensor ordering, and numerical precision used here.

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